

ELEC5305 Project Proposal

1. Project Title

Comparison of Acoustic Features for Male and Female Speaker Classification Using MATLAB

2. Student Information

Full Name: Meiqi Song

Student ID: 550200770

GitHub Username:meiqi0747

GitHub Project Link: <https://github.com/meiqi0747/elec5305-project-550200770>

GitHub Pages Link: <https://meiqi0747.github.io/elec5305-project-550200770/>

3. Project Overview

This project will investigate whether different audio features can be used to classify speech recordings labelled as male or female speakers. Speech signals contain many useful acoustic characteristics, such as pitch and frequency information, and these features can be extracted and analysed using MATLAB.

The project will mainly compare three feature sets. The first one will use Fundamental Frequency (F0), which is related to the pitch of the speaker's voice. The second will use Mel-Frequency Cepstral Coefficients (MFCCs), which provide information about the frequency characteristics of speech. The third will combine F0, MFCCs and Spectral Centroid. Two common classification methods, K-Nearest Neighbour (KNN) and Support Vector Machine (SVM), will then be tested using these features.

The aim is to find out which feature set gives better classification performance and whether combining different audio features can improve the result.

4. Background and Motivation

Speech contains more information than just the words being spoken. Different speakers can have different pitch ranges, frequency distributions and voice

characteristics. These differences can be measured using signal processing methods and may be useful for speaker classification.

Fundamental Frequency (F0) is related to voice pitch, but pitch alone may not be enough because different speakers can have overlapping ranges. MFCCs describe the short-term spectral shape of speech and provide more detailed information, while Spectral Centroid shows how energy is distributed across frequencies.

I chose this topic because it relates directly to audio feature extraction in ELEC5305 and can be implemented in MATLAB. Comparing these features can show which acoustic information is more useful for classification.

5. Proposed Methodology

The project will be completed mainly in MATLAB. A public speech dataset containing recordings with male and female speaker labels will be selected. A similar number of speakers from both groups will be used where possible so that the dataset is reasonably balanced.

Before extracting the features, the audio recordings will be pre-processed. Different recordings may have different sampling rates or amplitudes, so they will first be converted to a common sampling frequency and normalised. The speech will then be divided into short frames so that the features can be calculated over time.

Three feature conditions will be compared. The first condition will use F0 information only. For each recording, values such as mean F0, median F0 and the standard deviation of F0 can be calculated. The second condition will use MFCCs. The MFCC values from each recording will also be represented using simple statistics such as their mean and standard deviation. The final condition will combine F0, MFCCs and Spectral Centroid to see whether using several types of information together gives a better result.

KNN and SVM will be used as the two classification methods. MATLAB already provides functions for these classifiers, which means the project can focus more on feature comparison and result analysis rather than implementing the classifiers from the beginning.

To make the evaluation more reliable, recordings from the same speaker should not be divided between both training and testing data. This is important because otherwise the classifier may simply learn the characteristics of a particular speaker. A speaker-independent cross-validation method will therefore be used.

The performance of each experiment will be compared using classification accuracy. Precision, recall and F1-score will also be calculated, and confusion matrices will be used to show where incorrect classifications occur. The results from the different feature sets and classifiers will then be compared.

6. Expected Outcomes

The expected result of this project is a working MATLAB program that can extract acoustic features from speech recordings and classify them using KNN and SVM.

The main result will be a comparison between F0, MFCCs and the combined feature set. It is expected that F0 will provide a useful starting point, while MFCCs may give better results because they contain more information about the speech spectrum. The combined feature set may improve the result further, although this will depend on the experimental results.

The project will also produce several figures, including feature distributions, classification accuracy comparisons and confusion matrices. These results will be used to discuss why some methods perform better than others and what factors may cause classification errors.

7. Timeline

Weeks 1–2: Finalise the project topic, research question and GitHub project site.

Weeks 3–5: Conduct the literature review and select and prepare the speech dataset.

Weeks 6–7: Implement audio preprocessing and extract F0, MFCC and Spectral Centroid features.

Weeks 8–9: Implement KNN and SVM classification and five-fold cross-validation.

Weeks 10–11: Evaluate classification performance, generate figures and analyse classification errors.

Weeks 12–13: Complete the final report, GitHub documentation and project demonstration.

8. References

LEE, K.-H., KANG, S.-I., KIM, D.-H., & CHANG, J.-H. (2008). A Support Vector Machine-Based Gender Identification Using Speech Signal. *IEICE Transactions on Communications*, *E91.B*(10), 3326–3329. <https://doi.org/10.1093/ietcom/e91-b.10.3326>

Barkana, B. D., & Zhou, J. (2015). A new pitch-range based feature set for a speaker's age and gender classification. *Applied Acoustics*, *98*, 52–61. <https://doi.org/10.1016/j.apacoust.2015.04.013>

Li, M., Han, K. J., & Narayanan, S. (2013). Automatic speaker age and gender recognition using acoustic and prosodic level information fusion. *Computer Speech & Language*, *27*(1), 151–167. <https://doi.org/10.1016/j.csl.2012.01.008>

Yücesoy, E. (2024). Gender Recognition from Speech Signal Using CNN, KNN, SVM and RF. *Procedia Computer Science*, *235*, 2251–2257. <https://doi.org/10.1016/j.procs.2024.04.213>

Ardila, R., Branson, M., Davis, K., Henretty, M., Kohler, M., Meyer, J., Morais, R., Saunders, L., Tyers, F. M., & Weber, G. (2020). Common Voice: A Massively-Multilingual Speech Corpus. *arXiv.Org*.